KITABU QUEST

S4

CORE MATHEMATICS

Mountain Gorilla

Cover scene

students solving mathematics with graphs and models with a mountain gorilla mascot

- Clear lessons
- Worked examples
- Fun activities
- Real-life problems

GRADE 4

RWA REB-CBC aligned draft

Think - Explore - Achieve

Title Page

KITABU QUEST S4 Core Mathematics

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S4 learners study Core Mathematics one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use models, diagrams, algebra, graphs, and worked examples before independent practice.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Fundamentals of trigonometry
2. Propositional and predicate logic
3. Binary operations
4. Set of real numbers
5. Linear equations and inequalities
6. Quadratic equations and inequalities
7. Polynomial, rational and irrational functions
8. Limits of polynomial, rational and irrational functions
9. Differentiation of polynomials, rational and irrational functions and their applications
10. Vector spaces of real numbers
11. Concepts and operations on linear transformations in 2D
12. Matrices of and determinants of order
13. Points, straight lines and circles in 2D
14. Measures of dispersion
15. Combinatorics
16. Elementary probability

As you finish each topic, return to the success criteria and correct one answer before moving on.

Fundamentals of trigonometry: Lesson Opener

Learning focus: Fundamentals of trigonometry

Country link: a Rwandan school, market, farming, building, transport, or data problem for Fundamentals of trigonometry.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- use trigonometric circle and identities to determine trigonometric ratios and apply them to solve related problems
- define sine, cosine, and tangent (cosecant, secant and cotangent) of any angle - know special
- convert radians to degree and vice versa
- differentiate between complementary angles, supplementary angles and co- terminal angles
- represent graphically sine, cosine and tangent, functions and, together with the unit circle, use to relate values of any angle to the value for a positive acute angle
- use trigonometry, including the sine and cosine rules, to solve problems involving triangles

Inquiry questions:
- How can fundamentals of trigonometry help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Fundamentals of trigonometry: Learn

Fundamentals of trigonometry teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- Fundamentals of trigonometry
- Fundamentals
- trigonometry
- method
- model
- unit
- answer
- check

Fundamentals of trigonometry: Worked Example

Worked example for Fundamentals of trigonometry: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Fundamentals of trigonometry: Vocabulary And Study Skill

Use these words and study moves before you practise Fundamentals of trigonometry. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Fundamentals of trigonometry
- Fundamentals
- trigonometry
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Fundamentals of trigonometry without a clear example, method, or evidence.

Fundamentals of trigonometry: Guided Activity

Guided activity for Fundamentals of trigonometry:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Fundamentals of trigonometry: Practice And Correction

Practice for Fundamentals of trigonometry:

Main outcome: use trigonometric circle and identities to determine trigonometric ratios and apply them to solve related problems.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to fundamentals of trigonometry and solve it.

Fundamentals of trigonometry: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Fundamentals of trigonometry.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Fundamentals of trigonometry: Outcome Check

Outcome check for Fundamentals of trigonometry: Use trigonometric circle and identities to determine trigonometric ratios and apply them to solve related problems.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Fundamentals of trigonometry: Outcome Check

Outcome check for Fundamentals of trigonometry: Define sine, cosine, and tangent (cosecant, secant and cotangent) of any angle - know special.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Propositional and predicate logic: Lesson Opener

Learning focus: Propositional and predicate logic

Country link: a Rwandan school, market, farming, building, transport, or data problem for Propositional and predicate logic.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- use mathematical logic to organise scientific knowledge and as a tool of reasoning and argumentation in daily life
- distinguish between statement and proposition
- convert into logical formula composite propositions and vice versa
- draw the truth table of a composite proposition
- recognize the most often used tautologies
- (E g. De Morgan's Laws)

Inquiry questions:
- How can propositional and predicate logic help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Propositional and predicate logic: Learn

Propositional and predicate logic teaches how to collect, organize, read, and explain information.

Core idea:
- Data must answer a clear question.
- Tables, tally charts, pictographs, and bar charts make data easier to read.
- A good conclusion says what the data shows.

Key words:
- Propositional and predicate logic
- Propositional
- predicate
- logic
- method
- model
- unit
- answer

Propositional and predicate logic: Worked Example

Worked example for Propositional and predicate logic: A class records favourite fruits: mango 8, banana 12, pineapple 6. Which fruit is most popular?

1. Read the table values.
2. Compare 8, 12, and 6.
3. 12 is the greatest number.

Answer: Banana is most popular.

Propositional and predicate logic: Vocabulary And Study Skill

Use these words and study moves before you practise Propositional and predicate logic. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Propositional and predicate logic
- Propositional
- predicate
- logic
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Propositional and predicate logic without a clear example, method, or evidence.

Propositional and predicate logic: Guided Activity

Guided activity for Propositional and predicate logic:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Propositional and predicate logic: Practice And Correction

Practice for Propositional and predicate logic:

Main outcome: use mathematical logic to organise scientific knowledge and as a tool of reasoning and argumentation in daily life.

1. Ask 10 learners a simple question and make a tally table.
2. Draw a bar chart from the tally table.
3. Write one conclusion from the data.

Answer check:
Answer check: totals in the tally table and bar chart should match.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to propositional and predicate logic and solve it.

Propositional and predicate logic: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Propositional and predicate logic.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Propositional and predicate logic: Outcome Check

Outcome check for Propositional and predicate logic: Use mathematical logic to organise scientific knowledge and as a tool of reasoning and argumentation in daily life.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Propositional and predicate logic: Outcome Check

Outcome check for Propositional and predicate logic: Distinguish between statement and proposition.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Binary operations: Lesson Opener

Learning focus: Binary operations

Country link: a Rwandan school, market, farming, building, transport, or data problem for Binary operations.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- use mathematical logic to understand and perform operations using the properties of algebraic structures
- define a group, a ring , an integral domain and a field
- demonstrate that a set is (or is not) a group, a ring or a field under given operations
- demonstrate that a subset of a group is (or is not) a sub group
- determine the properties of a given binary operation
- formulate, using adequate symbols, a property of a binary operation and its negation

Inquiry questions:
- How can binary operations help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Binary operations: Learn

Binary operations teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- Binary operations
- Binary
- operations
- method
- model
- unit
- answer
- check

Binary operations: Worked Example

Worked example for Binary operations: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Binary operations: Vocabulary And Study Skill

Use these words and study moves before you practise Binary operations. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Binary operations
- Binary
- operations
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Binary operations without a clear example, method, or evidence.

Binary operations: Guided Activity

Guided activity for Binary operations:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Binary operations: Practice And Correction

Practice for Binary operations:

Main outcome: use mathematical logic to understand and perform operations using the properties of algebraic structures.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to binary operations and solve it.

Binary operations: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Binary operations.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Binary operations: Outcome Check

Outcome check for Binary operations: Use mathematical logic to understand and perform operations using the properties of algebraic structures.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Binary operations: Outcome Check

Outcome check for Binary operations: Define a group, a ring , an integral domain and a field.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Set of real numbers: Lesson Opener

Learning focus: Set of real numbers

Country link: a Rwandan school, market, farming, building, transport, or data problem for Set of real numbers.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- think critically using mathematical logic to understand and perform operations on the set of real numbers and its subsets using the properties of algebraic structures
- match a number and the set to which it belongs
- define a power, an exponential, a radical, a logarithm, the absolute value of a real number
- classify numbers into naturals, integers, rational and irrationals
- determine the restrictions on the variables in rational and irrational expressions
- illustrate each property of a power, an exponential, a radical, a logarithm, the absolute value of a real number

Inquiry questions:
- How can set of real numbers help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Set of real numbers: Learn

Set of real numbers teaches numbers below, at, and above zero.

Core idea:
- Positive numbers are greater than zero.
- Negative numbers are less than zero.
- Zero is the turning point.
- A number line helps you compare and order integers.

Rwanda example: temperature changes, money owed, points gained or lost in a game, and height above or below a reference point can use integers.

Key words:
- Set of real numbers
- real
- numbers
- method
- model
- unit
- answer
- check

Set of real numbers: Worked Example

Worked example for Set of real numbers: The morning temperature is 3 degrees above zero. At night it drops by 7 degrees. What is the night temperature?

1. Start at +3 on a number line.
2. Move 7 steps left because the temperature drops.
3. +3 - 7 = -4.

Answer: 4 degrees below zero.

Set of real numbers: Vocabulary And Study Skill

Use these words and study moves before you practise Set of real numbers. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Set of real numbers
- real
- numbers
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Set of real numbers without a clear example, method, or evidence.

Set of real numbers: Guided Activity

Guided activity for Set of real numbers:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Set of real numbers: Practice And Correction

Practice for Set of real numbers:

Main outcome: think critically using mathematical logic to understand and perform operations on the set of real numbers and its subsets using the properties of algebraic structures.

1. Solve one set of real numbers question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to set of real numbers and solve it.

Set of real numbers: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Set of real numbers.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Set of real numbers: Outcome Check

Outcome check for Set of real numbers: Think critically using mathematical logic to understand and perform operations on the set of real numbers and its subsets using the properties of algebraic structures.

1. Place -3, 0, and +4 on a number line.
2. Which number is greatest?
3. Find the distance from -3 to +4.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Set of real numbers: Outcome Check

Outcome check for Set of real numbers: Match a number and the set to which it belongs.

1. Place -3, 0, and +4 on a number line.
2. Which number is greatest?
3. Find the distance from -3 to +4.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Linear equations and inequalities: Lesson Opener

Learning focus: Linear equations and inequalities

Country link: a Rwandan school, market, farming, building, transport, or data problem for Linear equations and inequalities.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- model and solve algebraically or graphically daily life problems using linear equations or inequalities
- list and clarify the steps in modeling a problem by linear equations and inequalities
- solve equations and inequalities in one unknown
- solve parametric equations and inequalities in one unknown
- solve simultaneous equations in two unknowns. o Use equations in soling mathematical problems involving supply and demand, linear motions, electric circuitsalancing equations)
- appreciate, value and care for situations involving to quadratic equations and quadratic inequalities in daily life situation

Inquiry questions:
- How can linear equations and inequalities help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Linear equations and inequalities: Learn

Linear equations and inequalities teaches how lines and angles describe shapes and directions.

Core idea:
- A line can be straight, parallel, perpendicular, horizontal, vertical, or slanting.
- An angle is a turn between two lines.
- Draw and label diagrams carefully before measuring or classifying.

Key words:
- Linear equations and inequalities
- Linear
- equations
- inequalities
- method
- model
- unit
- answer

Linear equations and inequalities: Worked Example

Worked example for Linear equations and inequalities: A shape has one right angle. How can you show it clearly?

1. Draw the two meeting lines.
2. Mark the square corner to show a right angle.
3. Label it 90 degrees.
4. Name a classroom example, such as a book corner.

Answer: A right angle is a quarter turn, 90 degrees.

Linear equations and inequalities: Vocabulary And Study Skill

Use these words and study moves before you practise Linear equations and inequalities. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Linear equations and inequalities
- Linear
- equations
- inequalities
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Linear equations and inequalities without a clear example, method, or evidence.

Linear equations and inequalities: Guided Activity

Guided activity for Linear equations and inequalities:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Linear equations and inequalities: Practice And Correction

Practice for Linear equations and inequalities:

Main outcome: model and solve algebraically or graphically daily life problems using linear equations or inequalities.

1. Draw one right angle and label it 90 degrees.
2. Name one pair of parallel lines in the classroom.
3. Use a ruler to draw a straight line segment and label its endpoints.

Answer check:
Answer check: the right angle should show a square corner; parallel lines stay the same distance apart.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to linear equations and inequalities and solve it.

Linear equations and inequalities: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Linear equations and inequalities.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Linear equations and inequalities: Outcome Check

Outcome check for Linear equations and inequalities: Model and solve algebraically or graphically daily life problems using linear equations or inequalities.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Linear equations and inequalities: Outcome Check

Outcome check for Linear equations and inequalities: List and clarify the steps in modeling a problem by linear equations and inequalities.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Quadratic equations and inequalities: Lesson Opener

Learning focus: Quadratic equations and inequalities

Country link: a Rwandan school, market, farming, building, transport, or data problem for Quadratic equations and inequalities.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- model and solve algebraically or graphically daily life problems using quadratic equations or inequalities
- define a quadratic equation
- be able to solve problems related to quadratic equations
- apply critical thinking by solving any situation related to quadratic equations (economics problems, finance problems)
- determine the sign, sum and product of a quadratic equation
- discuss the parameter of a quadratic equation and a quadratic inequality

Inquiry questions:
- How can quadratic equations and inequalities help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Quadratic equations and inequalities: Learn

Quadratic equations and inequalities teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:
- Quadratic equations and inequalities
- Quadratic
- equations
- inequalities
- method
- model
- unit
- answer

Quadratic equations and inequalities: Worked Example

Worked example for Quadratic equations and inequalities: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:
1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

Quadratic equations and inequalities: Vocabulary And Study Skill

Use these words and study moves before you practise Quadratic equations and inequalities. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Quadratic equations and inequalities
- Quadratic
- equations
- inequalities
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Quadratic equations and inequalities without a clear example, method, or evidence.

Quadratic equations and inequalities: Guided Activity

Guided activity for Quadratic equations and inequalities:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Quadratic equations and inequalities: Practice And Correction

Practice for Quadratic equations and inequalities:

Main outcome: model and solve algebraically or graphically daily life problems using quadratic equations or inequalities.

1. Solve one quadratic equations and inequalities question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:
Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to quadratic equations and inequalities and solve it.

Quadratic equations and inequalities: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Quadratic equations and inequalities.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Quadratic equations and inequalities: Outcome Check

Outcome check for Quadratic equations and inequalities: Model and solve algebraically or graphically daily life problems using quadratic equations or inequalities.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Quadratic equations and inequalities: Outcome Check

Outcome check for Quadratic equations and inequalities: Define a quadratic equation.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Polynomial, rational and irrational functions: Lesson Opener

Learning focus: Polynomial, rational and irrational functions

Country link: a Rwandan school, market, farming, building, transport, or data problem for Polynomial, rational and irrational functions.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- use concepts and definitions of functions to determine the domain of rational functions and represent them graphically in simple cases and solve related problems
- identify a function as a rule and recognise rules that are not functions
- determine the domain and range of a function
- construct compositions of functions
- find whether a function is even, odd, or neither
- demonstrate an understanding of operations on polynomials, rational and irrational functions, and find the composite of two functions

Inquiry questions:
- How can polynomial, rational and irrational functions help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Polynomial, rational and irrational functions: Learn

Polynomial, rational and irrational functions teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- Polynomial, rational and irrational functions
- Polynomial
- rational
- irrational
- functions
- method
- model
- unit

Polynomial, rational and irrational functions: Worked Example

Worked example for Polynomial, rational and irrational functions: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Polynomial, rational and irrational functions: Vocabulary And Study Skill

Use these words and study moves before you practise Polynomial, rational and irrational functions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Polynomial, rational and irrational functions
- Polynomial
- rational
- irrational
- functions
- method
- model
- unit

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Polynomial, rational and irrational functions without a clear example, method, or evidence.

Polynomial, rational and irrational functions: Guided Activity

Guided activity for Polynomial, rational and irrational functions:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Polynomial, rational and irrational functions: Practice And Correction

Practice for Polynomial, rational and irrational functions:

Main outcome: use concepts and definitions of functions to determine the domain of rational functions and represent them graphically in simple cases and solve related problems.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to polynomial, rational and irrational functions and solve it.

Polynomial, rational and irrational functions: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Polynomial, rational and irrational functions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Polynomial, rational and irrational functions: Outcome Check

Outcome check for Polynomial, rational and irrational functions: Use concepts and definitions of functions to determine the domain of rational functions and represent them graphically in simple cases and solve related problems.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Polynomial, rational and irrational functions: Outcome Check

Outcome check for Polynomial, rational and irrational functions: Identify a function as a rule and recognise rules that are not functions.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Limits of polynomial, rational and irrational functions: Lesson Opener

Learning focus: Limits of polynomial, rational and irrational functions

Country link: a Rwandan school, market, farming, building, transport, or data problem for Limits of polynomial, rational and irrational functions.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- evaluate correctly limits of functions and apply them to solve related problems
- define the concept of limit for real-valued functions of one real variable
- evaluate the limit of a function and extend this concept to determine the asymptotes of the given function
- calculate limits of certain elementary functions
- develop introductory calculus reasoning
- solve problems involving continuity

Inquiry questions:
- How can limits of polynomial, rational and irrational functions help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Limits of polynomial, rational and irrational functions: Learn

Limits of polynomial, rational and irrational functions teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- Limits of polynomial, rational and irrational functions
- Limits
- polynomial
- rational
- irrational
- functions
- method
- model

Limits of polynomial, rational and irrational functions: Worked Example

Worked example for Limits of polynomial, rational and irrational functions: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Limits of polynomial, rational and irrational functions: Vocabulary And Study Skill

Use these words and study moves before you practise Limits of polynomial, rational and irrational functions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Limits of polynomial, rational and irrational functions
- Limits
- polynomial
- rational
- irrational
- functions
- method
- model

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Limits of polynomial, rational and irrational functions without a clear example, method, or evidence.

Limits of polynomial, rational and irrational functions: Guided Activity

Guided activity for Limits of polynomial, rational and irrational functions:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Limits of polynomial, rational and irrational functions: Practice And Correction

Practice for Limits of polynomial, rational and irrational functions:

Main outcome: evaluate correctly limits of functions and apply them to solve related problems.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to limits of polynomial, rational and irrational functions and solve it.

Limits of polynomial, rational and irrational functions: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Limits of polynomial, rational and irrational functions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Limits of polynomial, rational and irrational functions: Outcome Check

Outcome check for Limits of polynomial, rational and irrational functions: Evaluate correctly limits of functions and apply them to solve related problems.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Limits of polynomial, rational and irrational functions: Outcome Check

Outcome check for Limits of polynomial, rational and irrational functions: Define the concept of limit for real-valued functions of one real variable.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Differentiation of polynomials, rational and irrational functions and their applications: Lesson Opener

Learning focus: Differentiation of polynomials, rational and irrational functions and their applications

Country link: a Rwandan school, market, farming, building, transport, or data problem for Differentiation of polynomials, rational and irrational functions and their applications.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- use the gradient of a straight line as a measure of rate of change and apply this to tangent and normal of curves in various contexts and use the concepts of differentiation to solve and interpret related rates and optimisation problems in various contexts
- evaluate derivatives of functions using the definition of derivative
- define and evaluate from first principles the gradient at a point
- distinguish between techniques of differentiation to use in an appropriate context
- use properties of derivatives to differentiate polynomial, rational and irrational functions
- use first principles to determine the gradient of the tangent line to a curve at a point

Inquiry questions:
- How can differentiation of polynomials, rational and irrational functions and their applications help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Differentiation of polynomials, rational and irrational functions and their applications: Learn

Differentiation of polynomials, rational and irrational functions and their applications teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- Differentiation of polynomials, rational and irrational functions and their applications
- Differentiation
- polynomials
- rational
- irrational
- functions
- their
- applications

Differentiation of polynomials, rational and irrational functions and their applications: Worked Example

Worked example for Differentiation of polynomials, rational and irrational functions and their applications: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Differentiation of polynomials, rational and irrational functions and their applications: Vocabulary And Study Skill

Use these words and study moves before you practise Differentiation of polynomials, rational and irrational functions and their applications. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Differentiation of polynomials, rational and irrational functions and their applications
- Differentiation
- polynomials
- rational
- irrational
- functions
- their
- applications

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Differentiation of polynomials, rational and irrational functions and their applications without a clear example, method, or evidence.

Differentiation of polynomials, rational and irrational functions and their applications: Guided Activity

Guided activity for Differentiation of polynomials, rational and irrational functions and their applications:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Differentiation of polynomials, rational and irrational functions and their applications: Practice And Correction

Practice for Differentiation of polynomials, rational and irrational functions and their applications:

Main outcome: use the gradient of a straight line as a measure of rate of change and apply this to tangent and normal of curves in various contexts and use the concepts of differentiation to solve and interpret related rates and optimisation problems in various contexts.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to differentiation of polynomials, rational and irrational functions and their applications and solve it.

Differentiation of polynomials, rational and irrational functions and their applications: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Differentiation of polynomials, rational and irrational functions and their applications.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Differentiation of polynomials, rational and irrational functions and their applications: Outcome Check

Outcome check for Differentiation of polynomials, rational and irrational functions and their applications: Use the gradient of a straight line as a measure of rate of change and apply this to tangent and normal of curves in various contexts and use the concepts of differentiation to solve and interpret related rates and optimisation problems in various contexts.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Differentiation of polynomials, rational and irrational functions and their applications: Outcome Check

Outcome check for Differentiation of polynomials, rational and irrational functions and their applications: Evaluate derivatives of functions using the definition of derivative.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Vector spaces of real numbers: Lesson Opener

Learning focus: Vector spaces of real numbers

Country link: a Rwandan school, market, farming, building, transport, or data problem for Vector spaces of real numbers.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- determine the magnitude and angle between two vectors and be able to plot these vectors and point out dot product of two vectors
- define the scalar product of two vectors
- give examples of scalar product
- determine the magnitude of a vector and angle between two vectors
- calculate the scalar product of two vectors
- analyse a vector in term of size

Inquiry questions:
- How can vector spaces of real numbers help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Vector spaces of real numbers: Learn

Vector spaces of real numbers teaches how lines and angles describe shapes and directions.

Core idea:
- A line can be straight, parallel, perpendicular, horizontal, vertical, or slanting.
- An angle is a turn between two lines.
- Draw and label diagrams carefully before measuring or classifying.

Key words:
- Vector spaces of real numbers
- Vector
- spaces
- real
- numbers
- method
- model
- unit

Vector spaces of real numbers: Worked Example

Worked example for Vector spaces of real numbers: A shape has one right angle. How can you show it clearly?

1. Draw the two meeting lines.
2. Mark the square corner to show a right angle.
3. Label it 90 degrees.
4. Name a classroom example, such as a book corner.

Answer: A right angle is a quarter turn, 90 degrees.

Vector spaces of real numbers: Vocabulary And Study Skill

Use these words and study moves before you practise Vector spaces of real numbers. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Vector spaces of real numbers
- Vector
- spaces
- real
- numbers
- method
- model

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Vector spaces of real numbers without a clear example, method, or evidence.

Vector spaces of real numbers: Guided Activity

Guided activity for Vector spaces of real numbers:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Vector spaces of real numbers: Practice And Correction

Practice for Vector spaces of real numbers:

Main outcome: determine the magnitude and angle between two vectors and be able to plot these vectors and point out dot product of two vectors.

1. Draw one right angle and label it 90 degrees.
2. Name one pair of parallel lines in the classroom.
3. Use a ruler to draw a straight line segment and label its endpoints.

Answer check:
Answer check: the right angle should show a square corner; parallel lines stay the same distance apart.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to vector spaces of real numbers and solve it.

Vector spaces of real numbers: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Vector spaces of real numbers.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Vector spaces of real numbers: Outcome Check

Outcome check for Vector spaces of real numbers: Determine the magnitude and angle between two vectors and be able to plot these vectors and point out dot product of two vectors.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Vector spaces of real numbers: Outcome Check

Outcome check for Vector spaces of real numbers: Define the scalar product of two vectors.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Concepts and operations on linear transformations in 2D: Lesson Opener

Learning focus: Concepts and operations on linear transformations in 2D

Country link: a Rwandan school, market, farming, building, transport, or data problem for Concepts and operations on linear transformations in 2D.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- determine whether a transformation of 2IR is linear or not and perform operations on linear transformations
- define and distinguish between linear transformations in 2D
- define central symmetry, orthogonal projection of a vector, identical transformation
- define a rotation through: an angle about the origin.reflection in the x axis in y axis, in the line yx□
- show that a linear transformation is isomorphism in 2D or not
- perform operations on linear transformations in 2D

Inquiry questions:
- How can concepts and operations on linear transformations in 2d help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Concepts and operations on linear transformations in 2D: Learn

Concepts and operations on linear transformations in 2D teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- Concepts and operations on linear transformations in 2D
- Concepts
- operations
- linear
- transformations
- method
- model
- unit

Concepts and operations on linear transformations in 2D: Worked Example

Worked example for Concepts and operations on linear transformations in 2D: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Concepts and operations on linear transformations in 2D: Vocabulary And Study Skill

Use these words and study moves before you practise Concepts and operations on linear transformations in 2D. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Concepts and operations on linear transformations in 2D
- Concepts
- operations
- linear
- transformations
- method
- model
- unit

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Concepts and operations on linear transformations in 2D without a clear example, method, or evidence.

Concepts and operations on linear transformations in 2D: Guided Activity

Guided activity for Concepts and operations on linear transformations in 2D:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Concepts and operations on linear transformations in 2D: Practice And Correction

Practice for Concepts and operations on linear transformations in 2D:

Main outcome: determine whether a transformation of 2IR is linear or not and perform operations on linear transformations.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to concepts and operations on linear transformations in 2d and solve it.

Concepts and operations on linear transformations in 2D: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Concepts and operations on linear transformations in 2D.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Concepts and operations on linear transformations in 2D: Outcome Check

Outcome check for Concepts and operations on linear transformations in 2D: Determine whether a transformation of 2IR is linear or not and perform operations on linear transformations.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Concepts and operations on linear transformations in 2D: Outcome Check

Outcome check for Concepts and operations on linear transformations in 2D: Define and distinguish between linear transformations in 2D.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Matrices of and determinants of order: Lesson Opener

Learning focus: Matrices of and determinants of order

Country link: a Rwandan school, market, farming, building, transport, or data problem for Matrices of and determinants of order.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- use matrices and determinants of order 2 to solve systems of linear equations and to define transformations of 2D
- define the order of a matrix
- define a linear transformation in 2D by a matrix
- define operations on matrices of order 2
- show that a square matrix of order 2 is invertible or not
- reorganise data into matrices

Inquiry questions:
- How can matrices of and determinants of order 2 help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Matrices of and determinants of order: Learn

Matrices of and determinants of order teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- Matrices of and determinants of order
- Matrices
- determinants
- order
- method
- model
- unit
- answer

Matrices of and determinants of order: Worked Example

Worked example for Matrices of and determinants of order: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Matrices of and determinants of order: Vocabulary And Study Skill

Use these words and study moves before you practise Matrices of and determinants of order. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Matrices of and determinants of order
- Matrices
- determinants
- order
- method
- model
- unit
- answer

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Matrices of and determinants of order without a clear example, method, or evidence.

Matrices of and determinants of order: Guided Activity

Guided activity for Matrices of and determinants of order:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Matrices of and determinants of order: Practice And Correction

Practice for Matrices of and determinants of order:

Main outcome: use matrices and determinants of order 2 to solve systems of linear equations and to define transformations of 2D.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to matrices of and determinants of order and solve it.

Matrices of and determinants of order: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Matrices of and determinants of order.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Matrices of and determinants of order: Outcome Check

Outcome check for Matrices of and determinants of order: Use matrices and determinants of order 2 to solve systems of linear equations and to define transformations of 2D.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Matrices of and determinants of order: Outcome Check

Outcome check for Matrices of and determinants of order: Define the order of a matrix.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Points, straight lines and circles in 2D: Lesson Opener

Learning focus: Points, straight lines and circles in 2D

Country link: a Rwandan school, market, farming, building, transport, or data problem for Points, straight lines and circles in 2D.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- determine algebraic representations of lines, straight lines and circles in 2D
- define the coordinate of a point in 2D
- define a straight line knowing its: o 2 points. o Direction vector. o Gradient
- represent a point and / or a vector in 2D - Calculate the distance between two points in 2D and the mid - point of a segment in 2D
- determine equations of a straight line (vector equation, parametric equation, Cartesian equation)
- apply knowledge to find the centre, radius, and diameter to find out the equation of a circle

Inquiry questions:
- How can points, straight lines and circles in 2d help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Points, straight lines and circles in 2D: Learn

Points, straight lines and circles in 2D teaches the parts and measurements of circles.

Core idea:
- The centre is the fixed middle point of a circle.
- A radius goes from the centre to the circle.
- A diameter passes through the centre and is twice the radius.
- Circumference is the distance around the circle.
- A labelled diagram prevents mixing up radius and diameter.

Key words:
- Points, straight lines and circles in 2D
- Points
- straight
- lines
- circles
- method
- model
- unit

Points, straight lines and circles in 2D: Worked Example

Worked example for Points, straight lines and circles in 2D: A circle has radius 7 cm. Find its diameter.

1. Diameter = 2 x radius.
2. Radius = 7 cm.
3. 2 x 7 cm = 14 cm.
4. Check the diagram: the diameter must pass through the centre.

Answer: The diameter is 14 cm.

Points, straight lines and circles in 2D: Vocabulary And Study Skill

Use these words and study moves before you practise Points, straight lines and circles in 2D. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Points, straight lines and circles in 2D
- Points
- straight
- lines
- circles
- method
- model
- unit

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Points, straight lines and circles in 2D without a clear example, method, or evidence.

Points, straight lines and circles in 2D: Guided Activity

Guided activity for Points, straight lines and circles in 2D:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Points, straight lines and circles in 2D: Practice And Correction

Practice for Points, straight lines and circles in 2D:

Main outcome: determine algebraic representations of lines, straight lines and circles in 2D.

1. Draw a circle and label the centre, radius, and diameter.
2. If the radius is 6 cm, find the diameter.
3. Name one circular object at school or home and describe its radius or diameter.

Answer check:
Answer: if radius = 6 cm, diameter = 12 cm. A correct diagram shows the radius from centre to circle and diameter through the centre.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to points, straight lines and circles in 2d and solve it.

Points, straight lines and circles in 2D: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Points, straight lines and circles in 2D.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Points, straight lines and circles in 2D: Outcome Check

Outcome check for Points, straight lines and circles in 2D: Determine algebraic representations of lines, straight lines and circles in 2D.

1. Draw and label a circle centre, radius, and diameter.
2. If radius is 4 cm, find diameter.
3. Explain why diameter is twice the radius.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Points, straight lines and circles in 2D: Outcome Check

Outcome check for Points, straight lines and circles in 2D: Define the coordinate of a point in 2D.

1. Draw and label a circle centre, radius, and diameter.
2. If radius is 4 cm, find diameter.
3. Explain why diameter is twice the radius.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Measures of dispersion: Lesson Opener

Learning focus: Measures of dispersion

Country link: a Rwandan school, market, farming, building, transport, or data problem for Measures of dispersion.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- extend understanding, analysis and interpretation of data arising from problems and questions in daily life to include the standard deviation
- define the variance, standard deviation and the coefficient of variation
- analyse and interpret critically data and infer conclusions
- determine the measures of dispersion of a given statistical series
- apply and explain the standard deviation as the more convenient measure of the variability in the interpretation of data
- express the coefficient of variation as a measure of the spread of a set of data as a proportion of its mean

Inquiry questions:
- How can measures of dispersion help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Measures of dispersion: Learn

Measures of dispersion teaches how two quantities change together at the same rate.

Core idea:
- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:
- Measures of dispersion
- Measures
- dispersion
- method
- model
- unit
- answer
- check

Measures of dispersion: Worked Example

Worked example for Measures of dispersion: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: 900 / 3 = 300 francs.
2. Multiply by 5 books: 300 x 5 = 1500 francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Measures of dispersion: Vocabulary And Study Skill

Use these words and study moves before you practise Measures of dispersion. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Measures of dispersion
- Measures
- dispersion
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Measures of dispersion without a clear example, method, or evidence.

Measures of dispersion: Guided Activity

Guided activity for Measures of dispersion:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Measures of dispersion: Practice And Correction

Practice for Measures of dispersion:

Main outcome: extend understanding, analysis and interpretation of data arising from problems and questions in daily life to include the standard deviation.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:
Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because 18 / 2 = 9 km per hour.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to measures of dispersion and solve it.

Measures of dispersion: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Measures of dispersion.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Measures of dispersion: Outcome Check

Outcome check for Measures of dispersion: Extend understanding, analysis and interpretation of data arising from problems and questions in daily life to include the standard deviation.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Measures of dispersion: Outcome Check

Outcome check for Measures of dispersion: Define the variance, standard deviation and the coefficient of variation.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Combinatorics: Lesson Opener

Learning focus: Combinatorics

Country link: a Rwandan school, market, farming, building, transport, or data problem for Combinatorics.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- use combinations and permutations to determine the number of ways a random experiment occurs
- define the combinatorial analysis
- recognise whether repetition is allowed or not, and if order matters or not in performing a given experiment
- construct Pascal's triangle
- distinguish between permutations and combinations
- determine the number of permutations and combinations of "n" items, "r" taken at a time

Inquiry questions:
- How can combinatorics help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Combinatorics: Learn

Combinatorics teaches how to read, write, convert, and use time.

Core idea:
- Time can be shown on analogue clocks, digital clocks, calendars, and timetables.
- 60 seconds make 1 minute, and 60 minutes make 1 hour.
- Timetables help you find start time, end time, and duration.

Key words:
- Combinatorics
- method
- model
- unit
- answer
- check

Combinatorics: Worked Example

Worked example for Combinatorics: A lesson starts at 8:10 and ends at 8:50. How long is the lesson?

1. Count from 8:10 to 8:50.
2. 10 minutes to 8:20, 20 more to 8:40, 10 more to 8:50.
3. Total time = 40 minutes.

Answer: 40 minutes.

Combinatorics: Vocabulary And Study Skill

Use these words and study moves before you practise Combinatorics. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Combinatorics
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Combinatorics without a clear example, method, or evidence.

Combinatorics: Guided Activity

Guided activity for Combinatorics:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Combinatorics: Practice And Correction

Practice for Combinatorics:

Main outcome: use combinations and permutations to determine the number of ways a random experiment occurs.

1. Read a clock showing 7:30 and write the time in words.
2. Convert 2 hours to minutes.
3. A trip starts at 9:15 and ends at 10:00. Find the duration.

Answer check:
Answers: 2 hours = 120 minutes. 9:15 to 10:00 is 45 minutes.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to combinatorics and solve it.

Combinatorics: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Combinatorics.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Combinatorics: Outcome Check

Outcome check for Combinatorics: Use combinations and permutations to determine the number of ways a random experiment occurs.

1. Convert 3 hours to minutes.
2. Find the duration from 8:20 to 9:05.
3. Write the answer in minutes.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Combinatorics: Outcome Check

Outcome check for Combinatorics: Define the combinatorial analysis.

1. Convert 3 hours to minutes.
2. Find the duration from 8:20 to 9:05.
3. Write the answer in minutes.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Elementary probability: Lesson Opener

Learning focus: Elementary probability

Country link: a Rwandan school, market, farming, building, transport, or data problem for Elementary probability.

Progression focus: prepare for assessment and real-world use through extended tasks and correction; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:
- use counting techniques and concepts of probability to determine the probability of possible outcomes of events occurring under equally likely assumptions
- define probability and explain probability as a measure of chance
- distinguish between mutually exclusive and non- exclusive events and compute their probabilities
- use and apply properties of probability to calculate the number of possible outcomes of occurring events under equally likely assumptions
- determine and explain expectations from an experiment with possible outcomes
- appreciate the use of probability as a measure of chance

Inquiry questions:
- How can elementary probability help you solve a real problem in Core Mathematics?

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Elementary probability: Learn

Elementary probability teaches how likely an event is.

Core idea:
- Some events are certain, likely, unlikely, or impossible.
- A simple experiment can help you compare chances.
- Record results before making a conclusion.

Key words:
- Elementary probability
- Elementary
- probability
- method
- model
- unit
- answer
- check

Elementary probability: Worked Example

Worked example for Elementary probability: A bag has 4 red counters and 1 blue counter. Which colour is more likely to be picked?

1. Compare the number of counters.
2. Red has 4 chances; blue has 1 chance.
3. More red counters means red is more likely.

Answer: Red is more likely.

Elementary probability: Vocabulary And Study Skill

Use these words and study moves before you practise Elementary probability. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Elementary probability
- Elementary
- probability
- method
- model
- unit
- answer
- check

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Elementary probability without a clear example, method, or evidence.

Elementary probability: Guided Activity

Guided activity for Elementary probability:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Elementary probability: Practice And Correction

Practice for Elementary probability:

Main outcome: use counting techniques and concepts of probability to determine the probability of possible outcomes of events occurring under equally likely assumptions.

1. Write one certain event, one likely event, and one impossible event.
2. Toss a coin 10 times and record heads/tails.
3. Use your results to say what happened more often.

Answer check:
Answer check: certain means must happen; impossible means cannot happen.

Correction check:
- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to elementary probability and solve it.

Elementary probability: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Elementary probability.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- method is clear
- working is shown step by step
- answer is checked in context

Elementary probability: Outcome Check

Outcome check for Elementary probability: Use counting techniques and concepts of probability to determine the probability of possible outcomes of events occurring under equally likely assumptions.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Elementary probability: Outcome Check

Outcome check for Elementary probability: Define probability and explain probability as a measure of chance.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:
- method is clear
- working is shown step by step
- answer is checked in context

Fundamentals of trigonometry: Review Clinic 1

Use this review clinic to strengthen Core Mathematics without copying earlier answers.

1. Choose one idea from Fundamentals of trigonometry that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Propositional and predicate logic: Review Clinic 2

Use this review clinic to strengthen Core Mathematics without copying earlier answers.

1. Choose one idea from Propositional and predicate logic that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Binary operations: Review Clinic 3

Use this review clinic to strengthen Core Mathematics without copying earlier answers.

1. Choose one idea from Binary operations that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Set of real numbers: Review Clinic 4

Use this review clinic to strengthen Core Mathematics without copying earlier answers.

1. Choose one idea from Set of real numbers that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Core Mathematics.

- Fundamentals of trigonometry: Explain this term using an example from the book, your school, or your community.
- Propositional and predicate logic: Explain this term using an example from the book, your school, or your community.
- Binary operations: Explain this term using an example from the book, your school, or your community.
- Set of real numbers: Explain this term using an example from the book, your school, or your community.
- Linear equations and inequalities: Explain this term using an example from the book, your school, or your community.
- Quadratic equations and inequalities: Explain this term using an example from the book, your school, or your community.
- Polynomial, rational and irrational functions: Explain this term using an example from the book, your school, or your community.
- Limits of polynomial, rational and irrational functions: Explain this term using an example from the book, your school, or your community.
- Differentiation of polynomials, rational and irrational functions and their applications: Explain this term using an example from the book, your school, or your community.
- Vector spaces of real numbers: Explain this term using an example from the book, your school, or your community.
- Concepts and operations on linear transformations in 2D: Explain this term using an example from the book, your school, or your community.
- Matrices of and determinants of order: Explain this term using an example from the book, your school, or your community.
- Points, straight lines and circles in 2D: Explain this term using an example from the book, your school, or your community.
- Measures of dispersion: Explain this term using an example from the book, your school, or your community.
- Combinatorics: Explain this term using an example from the book, your school, or your community.
- Elementary probability: Explain this term using an example from the book, your school, or your

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Core Mathematics answer shows the model, method, units or labels, final answer, and a check. Use the worked examples and answer checks in each practice page to correct your work. When an answer is wrong, find the first step that changed the meaning and correct that step before trying a new question.

Final Project

Create a final project for Core Mathematics that shows what you can now do independently.

Your project must include:
1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.